

CHANDIGARH ENGINEERING COLLEGE JHANNERI

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ENGINEERING CLINICS PROJECT SYNOPSIS

B. Tech - 3rd Semester

**“AI-Based Solar Power Generation Forecasting and
Panel Performance Anomaly Detection”**

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1. Abstract

“AI-Based Solar Power Generation Forecasting and Panel Performance Anomaly Detection”

Solar photovoltaic (PV) power generation is inherently intermittent and susceptible to environmental and physical degradation, such as dust buildup, localized shading, thermal stress, and electrical faults. Traditional solar installations lack localized intelligence, operating without predictive yield awareness or automated fault detection. This project presents an intelligent, budget-friendly edge-to-cloud IoT and Machine Learning system that combines short-term solar power forecasting with multi-tier real-time anomaly detection. The hardware architecture utilizes an Arduino Uno microcontroller interfaced with an INA219 high-side power sensor (monitoring bus voltage, current, and wattage), a DHT11 environmental sensor (measuring ambient temperature and humidity), and an analog Light Dependent Resistor (LDR) to gauge irradiance. Telemetry is streamed over a 115200 baud serial pipeline to a Python backend backed by SQLite time-series storage. A Random Forest regression model performs time-series power forecasting 1 to 6 hours ahead using solar ephemeris, atmospheric variables, and rolling lag features. Simultaneously, a three-layer anomaly detection framework—combining rolling Z-score statistical bounds, domain-specific physics rules (e.g., voltage drop under high irradiance, power-to-light ratio degradation), and an unsupervised Isolation Forest ML algorithm—detects operational anomalies in real time. A responsive Streamlit SCADA dashboard provides live electrical telemetry, interactive forecasting curves, an anomaly alert timeline, and a composite panel health score (0–100%), offering a complete, low-cost predictive maintenance solution for modern solar PV installations.

2. Introduction

The global transition toward renewable energy has established solar photovoltaic (PV) power as a cornerstone of decarbonization and smart grid initiatives. However, realizing the maximum operational efficiency and economic reliability of PV installations remains a formidable engineering challenge. Solar power generation is heavily dictated by fluctuating weather dynamics, solar geometry, atmospheric transmittance, and ambient temperatures. Moreover, installed solar panels regularly endure adverse environmental stressors—such as dust/soiling accumulation, partial shading from transient objects, wiring corrosion, cell micro-cracks, and overheating—which cumulatively degrade power output by 10% to 25% if left unaddressed.

In typical decentralized and rooftop PV installations, monitoring is either absent or restricted to passive energy meters that only record net power without predictive foresight or diagnostic capability. When power output drops, system operators cannot distinguish whether the attenuation is due to natural cloud cover or physical panel degradation. This absence of diagnostic intelligence leads to delayed maintenance, prolonged energy loss, and potential safety risks.

This project proposes an end-to-end smart solution: **“AI-Based Solar Power Generation Forecasting and Panel Performance Anomaly Detection.”** By fusing low-cost embedded sensing with state-of-the-art machine learning algorithms, the system achieves dual capabilities: accurately predicting future power generation and proactively detecting performance faults. This project directly aligns with Computer Science & Engineering domains, integrating embedded systems, time-series data pipelines, machine learning regression, unsupervised anomaly detection, and interactive web dashboard development.

3. Problem Statement

Current small-to-medium scale photovoltaic installations suffer from two critical operational deficiencies: the inability to forecast short-term energy yield and the lack of automated, localized fault and anomaly detection.

First, grid operators and consumers lack localized, short-term power forecasting. General meteorological forecasts lack micro-climate specificity for individual panel setups, making daily energy planning, battery storage scheduling, and load dispatching highly inefficient.

Second, traditional solar monitoring systems are purely reactive. Faults such as panel soiling (dust buildup), loose terminal connections, diode bypass failures, and severe thermal overheating often go unnoticed for weeks until manual physical inspections are conducted or electricity bills indicate substantial losses. Conventional simple threshold alarms fail because solar irradiance naturally varies throughout the day; a drop in power is normal at dusk or during clouds but abnormal under clear midday skies. Distinguishing between environmental changes and genuine hardware faults requires multivariate, context-aware machine learning.

There is an urgent requirement for a unified, low-cost computational system that correlates real-time electrical parameters (voltage, current, power) with ambient environmental indicators (irradiance, temperature, humidity) using machine learning to deliver accurate future generation forecasts and immediate, automated anomaly alerting.

4. Objectives

- **Hardware Data Acquisition:** To develop a low-cost, multi-sensor data acquisition hardware module utilizing an Arduino Uno, INA219 digital power sensor, DHT11 environmental sensor, and an analog LDR.
- **Time-Series Data Pipeline:** To implement a real-time data pipeline in Python backed by SQLite for robust local telemetry logging and feature extraction.
- **AI Power Forecasting:** To build a predictive model using Random Forest Regression capable of predicting PV power generation 1 to 6 hours ahead.
- **Tri-Layer Anomaly Detection:** To design an anomaly detection framework integrating rolling Z-score statistical boundaries, domain physics rules, and an unsupervised Isolation Forest algorithm.
- **Panel Health Scoring:** To establish an automated Panel Health Scoring algorithm (0–100%) that synthesizes anomaly severity into actionable maintenance recommendations.
- **SCADA Web Dashboard:** To develop a professional, responsive SCADA-style Streamlit web dashboard for live telemetry visualization, interactive forecasting graphs, anomaly log explorer, and dataset export.

5. Scope of the Project

The scope of this project encompasses the complete engineering cycle from embedded sensor integration to machine learning model deployment and supervisory control visualization.

- **Inclusions:** Includes full hardware circuit design, sensor wiring, embedded Arduino C/C++ firmware with Exponential Moving Average (EMA) filtering, USB serial data communication, time-series SQLite database architecture, Random Forest forecasting, multi-algorithm anomaly detection, and a 5-tab Streamlit dashboard.
- **Exclusions:** High-voltage commercial three-phase grid-tie inverter synchronization, industrial SCADA PLC integration, and physical drone-based thermal imaging are beyond the prototype scope, keeping the solution accessible and low-cost.
- **Operating Environment:** Designed for academic research, rooftop testbenches, and micro-grid installations under both outdoor sunlight and simulated fault conditions.
- **Target Beneficiaries:** Solar rooftop owners, commercial micro-grid managers, smart-grid researchers, and renewable energy facility operators seeking low-cost predictive maintenance and yield forecasting.

6. Literature Review / Existing System

Existing photovoltaic monitoring and predictive systems generally fall into three categories, each exhibiting distinct limitations:

1. Static Threshold & Manual Inspection Methods:

Conventional industrial installations rely on threshold-based SCADA alarms or manual periodic checks. These systems trigger warnings only when power falls below fixed thresholds. However, because solar generation varies dynamically from dawn to dusk and across seasons, static thresholds produce either rampant false alarms during cloudy spells or fail to catch gradual degradation caused by dust and cell deterioration.

2. Classical Statistical Forecasting (ARIMA/Physical Models):

Traditional numerical weather prediction (NWP) models and statistical time-series algorithms (e.g., ARIMA) have been applied to solar power forecasting. While useful for regional macro-forecasts, they fail to capture localized micro-climatic fluctuations (such as passing clouds or local building shadows) and require extensive cloud computing infrastructure, rendering them impractical for localized edge deployments.

3. Machine Learning & Anomaly Detection Advances:

Recent academic literature has demonstrated the efficacy of Deep Learning (LSTM, GRU) for solar forecasting. However, deep recurrent networks demand significant computational memory and training overhead. For embedded and edge setups, Random Forest Regressors have been proven to deliver competitive accuracy ($R^2 > 0.90$) with minimal training latency, built-in resistance to overfitting, and high explainability through feature importance rankings.

This proposed project bridges these gaps by combining lightweight edge computing (Arduino + Python) with a hybrid intelligence framework: a Random Forest Regressor for local generation forecasting and a 3-tier anomaly detector (Statistical Z-score + Physics rules + Isolation Forest) that contextualizes electrical performance against ambient environmental conditions in real time.

7. Methodology

The project follows a structured modular engineering methodology comprising four integrated phases:

- **Phase 1 - Embedded Data Acquisition:** Sensors (INA219, DHT11, LDR) are polled by an Arduino Uno every 2 seconds. Exponential Moving Average (EMA) filtering is applied to eliminate electrical noise before emitting structured JSON packets over USB serial.
- **Phase 2 - Telemetry Ingestion & Feature Engineering:** A multi-threaded Python reader parses JSON telemetry and appends timestamped records to a normalized SQLite database. Derived features—including hour angles, solar elevation via NOAA equations, rolling means, lag features, and power-to-light ratios—are computed continuously.
- **Phase 3 - Machine Learning Engine:** The Random Forest Regressor is trained on historical features to predict future power output at 15-minute intervals. Concurrently, new readings pass through Z-score filters, domain physics rules, and an unsupervised Isolation Forest model to detect faults.
- **Phase 4 - Supervisory Dashboard & Analytics:** The Streamlit SCADA dashboard updates live, visualizing real-time metrics, forecast trajectories with confidence metrics (MAE, RMSE, R^2), anomaly scatter plots, and composite health scores.

8. Tools and Technologies Used

- **Front-End / UI:** Streamlit (Python framework) for rapid, reactive SCADA dashboard rendering.
- **Visualization:** Plotly & Plotly Express for dark-theme interactive time-series charts, gauges, and anomaly scatter plots.
- **Back-End & ML:** Python 3.10+ with scikit-learn (Random Forest, Isolation Forest), pandas, and numpy.
- **Embedded Firmware:** C/C++ (Arduino Core) with Adafruit INA219 and bit-bang DHT11 driver.
- **Database:** SQLite 3 for relational, zero-configuration local time-series data storage.
- **Communication:** pyserial for robust bidirectional USB communication with the microcontroller.
- **IDEs & Tools:** VS Code, Arduino IDE 2.x, Git & GitHub for repository management and version control.

9. Hardware and Software Requirements

Hardware Requirements Specification Table:

Category	Component Specification	Qty
Microcontroller	Arduino Uno R3 (ATmega328P / CH340)	1
Power Sensing	INA219 I2C High-Side Voltage, Current & Power Sensor	1
Climate Sensing	DHT11 Digital Temperature & Humidity Sensor	1
Optical Sensing	Light Dependent Resistor (LDR, 5mm) + 10k Ω Resistor	1 set
PV Generation	5V - 6V / 1W - 2W Mini Polycrystalline Solar Panel	1
Prototyping	400-Point Solderless Breadboard	1
Interconnects	Male-to-Male & Male-to-Female Jumper Wires	1 Pack
Connectivity	USB Type-A to Type-B Interface Cable	1

Software Requirements:

- **Operating System:** Windows 10 / 11, macOS, or Ubuntu Linux (64-bit).
- **Development Environments (IDEs):** Visual Studio Code (VS Code) and Arduino IDE v2.x.
- **Programming Languages:** Python v3.10+ and C/C++ (Arduino AVR toolchain).
- **Python Libraries:** scikit-learn (RandomForestRegressor, IsolationForest), pandas, numpy, pyserial, streamlit, plotly.
- **Embedded Libraries:** Wire.h, Adafruit_INA219.h.

10. Outcome

The successful completion of this project delivers an end-to-end, functional AI-driven solar monitoring and predictive maintenance platform that integrates physical embedded hardware with advanced machine learning.

Key Deliverables and Achievements:

- **Fully Functional IoT Sensor Prototype:** A low-cost, plug-and-play sensing kit that streams precision voltage, current, power, temperature, humidity, and irradiance metrics at 0.5 Hz over serial.
- **Predictive Power Generation Forecasting:** A validated Random Forest regression model achieving high prediction accuracy ($R^2 > 0.85$ – 0.92) for 1 to 6-hour power output forecasting, providing essential visibility for load balancing and battery management.
- **Tri-Layer Anomaly Detection Engine:** A comprehensive anomaly engine that successfully isolates dust accumulation, partial panel shading, loose wiring contacts, and thermal overheating with zero false alarms under normal dawn-to-dusk transitions.
- **Dynamic Panel Health Scoring:** A real-time health indicator (0–100%) that translates multivariate anomalies into clear, human-understandable maintenance advisories (e.g., 'Dust cleanup required', 'Inspect wiring').
- **SCADA-Style Streamlit Dashboard:** A responsive, 5-tab web dashboard facilitating live telemetry tracking, forecast evaluation, historical anomaly investigation, and full dataset CSV export for academic research.

11. References

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